Department of Information Science & Engineering

IS3601-1 Problem Solving on Online Coding Platform

LAB SEMESTER END EXAMINATION – APRIL 2026

Part - A				
Sl. No	**LeetCode Number**	**Problem Description**	**Pattern**	**CO**
1	#303. Range Sum Query	Given an integer array nums, handle multiple queries of the following type: Calculate the **sum** of the elements of nums between indices left and right **inclusive** where left <= right.	**Prefix Sum**	**1, 2**
2	#560. Subarray Sum Equals K	Given an array of integers nums and an integer k, return *the total number of subarrays whose sum equals to* k.		
3	#125. Valid Palindrome	A phrase is a **palindrome** if, after converting all uppercase letters into lowercase letters and removing all non-alphanumeric characters, it reads the same forward and backward. Alphanumeric characters include letters and numbers. Given a string s, return true *if it is a **palindrome**, or* false *otherwise*.	**Two Pointers**	**1, 2**
4	#167. Two Sum II - Input Array Is Sorted	Given a **1-indexed** array of integers numbers that is already ***sorted in non-decreasing order***, find two numbers such that they add up to a specific target number. Let these two numbers be numbers[$index_1$] and numbers[$index_2$] where 1 <= $index_1$ < $index_2$ <= numbers.length. Return *the indices of the two numbers,* $index_1$ *and* $index_2$*, **added by one** as an integer array* [$index_1$, $index_2$] *of length 2.*		
5	#643. Maximum Average Subarray I	You are given an integer array nums consisting of n elements, and an integer k. Find a contiguous subarray whose **length is equal to** k that has the maximum average value and return *this value*.	**Sliding Window**	**1, 2**
6	#3. Longest Substring Without Repeating Characters	Given a string s, find the length of the **longest substring** without repeating characters.		
7	#141. Linked List Cycle	Given head, the head of a linked list, determine if the linked list has a cycle in it.	**Fast & Slow Pointers**	**1, 2**
8	#876. Middle of the Linked List	Given the head of a singly linked list, return *the middle node of the linked list*.		
9	#206. Reverse Linked List	Given the head of a singly linked list, reverse the list, and return *the reversed list*.	**LinkedList In-place Reversal**	**1, 2**
10	#92. Reverse Linked List II	Given the head of a singly linked list and two integers left and right where left <= right, reverse the nodes		

		of the list from position left to position right, and return *the reversed list*.		
11	#496. Next Greater Element I	The **next greater element** of some element x in an array is the **first greater** element that is **to the right** of x in the same array. You are given two **distinct 0-indexed** integer arrays nums1 and nums2, where nums1 is a subset of nums2. For each 0 <= i < nums1.length, find the index j such that nums1[i] == nums2[j] and determine the **next greater element** of nums2[j] in nums2. If there is no next greater element, then the answer for this query is -1. Return *an array* ans *of length* nums1.length *such that* ans[i] *is the* ***next greater element*** *as described above.*	**Monotonic Stack**	**1, 2**
12	#739. Daily Temperatures	Given an array of integers temperatures represents the daily temperatures, return *an array* answer *such that* answer[i] *is the number of days you have to wait after the* i^{th} *day to get a warmer temperature*. If there is no future day for which this is possible, keep answer[i] == 0 instead.		

Part - B				
Sl. No	**LeetCode Number**	**Problem Description**	**Pattern**	**CO**
1	#238. Product of Array Except Self	Given an integer array nums, return *an array* answer *such that* answer[i] *is equal to the product of all the elements of* nums *except* nums[i].	**Prefix Sum**	**1, 2**
2	#11. Container With Most Water	You are given an integer array height of length n. There are n vertical lines drawn such that the two endpoints of the ith line are (i, 0) and (i, height[i]). Find two lines that together with the x-axis form a container, such that the container contains the most water. Return *the maximum amount of water a container can store*.	**Two Pointers**	**1, 2**
3	#424. Longest Repeating Character Replacement	You are given a string s and an integer k. You can choose any character of the string and change it to any other uppercase English character. You can perform this operation at most k times. Return *the length of the longest substring containing the same letter you can get after performing the above operations*.	**Sliding Window**	**1, 2**

4	#19. Remove Nth Node From End of List	Given the head of a linked list, remove the nth node from the end of the list and return its head.	**Fast & Slow Pointers**	**1, 2**
5	#61. Rotate List	Given the head of a linked list, rotate the list to the right by k places.	**LinkedList In-place Reversal**	**1, 2**
6	#901. Online Stock Span	Design an algorithm that collects daily price quotes for some stock and returns **the span** of that stock's price for the current day. The **span** of the stock's price in one day is the maximum number of consecutive days (starting from that day and going backward) for which the stock price was less than or equal to the price of that day. • For example, if the prices of the stock in the last four days is [7,2,1,2] and the price of the stock today is 2, then the span of today is 4 because starting from today, the price of the stock was less than or equal 2 for 4 consecutive days. • Also, if the prices of the stock in the last four days is [7,34,1,2] and the price of the stock today is 8, then the span of today is 3 because starting from today, the price of the stock was less than or equal 8 for 3 consecutive days.	**Monotonic Stack**	**1, 2**

Course Outcomes

CO1: Approach and solve complex problems systematically, demonstrating improved logical thinking and algorithmic skills.

CO2: Demonstrate proficiency in at least one programming language, writing efficient and optimized code to solve various problems on online coding platforms.

MSE Scheme

Assessment	Weightage in Marks
Write Up	10
Execution	30
Viva Voce	10
Total	**50**